Percent Applications
Answer Key

Grade 6–7 Ratios and Proportional Relationships

Quick Answers

1. 30	4. 80	7. 200	10. 250
2. 84	5. 120	8. 20	11. 25
3. 30	6. 18	9. 6.75	12. 140

Common wrong answers

9. *If they got 1481.48: divided the whole by the part (80 over 5.40) instead of part over whole*

Every solution uses the same relationship, $\frac{\text{part}}{\text{whole}} = \frac{\text{percent}}{100}$. What changes from problem to problem is *which* of the three is missing. The first line of each solution names it.

1. 18 of 60 seats filled — what percent?

Part = 18, whole = 60, percent unknown.

$$\frac{18}{60} = \frac{p}{100} \quad \Rightarrow \quad 60p = 1800 \quad \Rightarrow \quad p = \frac{1800}{60}$$

Equivalently, $18 \div 60 = 0.3$ and 0.3×100 gives the rate. $\boxed{30\%}$

2. 21 of 25 correct — what percent?

Part = 21, whole = 25, percent unknown. Because $25 \times 4 = 100$, scale the whole ratio up by 4 instead of dividing:

$$\frac{21}{25} = \frac{21 \times 4}{25 \times 4} = \frac{84}{100}$$

A denominator of 100 reads off as the percent directly. $\boxed{84\%}$

3. 40% of the club is 12 members — how many members?

Here the *percent* and the *part* are known and the whole is missing, so the unknown goes in the denominator:

$$\frac{12}{n} = \frac{40}{100} \quad \Rightarrow \quad 40n = 1200 \quad \Rightarrow \quad n = 30$$

Check: 40% of 30 is $0.4 \times 30 = 12$ ✓. $\boxed{n = 30 \text{ members}}$

4. 36 of 45 penalties made — what percent?

Part = 36, whole = 45, percent unknown.

$$\frac{36}{45} = \frac{4}{5} = \frac{80}{100}$$

Simplifying first (36/45 divides by 9) turns this into a fifths problem, and fifths convert to hundredths by multiplying by 20. $\boxed{80\%}$

5. An 18-dollar discount is 15% of the original price — find the price.

The 18 dollars is the part; 15 is the percent; the original price is the missing whole.

$$\frac{18}{n} = \frac{15}{100} \quad \Rightarrow \quad 15n = 1800 \quad \Rightarrow \quad n = 120$$

Check: 15% of 120 is $0.15 \times 120 = 18$ ✓. $\boxed{n = 120 \text{ dollars}}$

6. 45 of 250 surveyed chose blue — what percent?

Part = 45, whole = 250, percent unknown. Since $250 \times 4 = 1000$, scaling by 4 lands on thousandths, so halve once more — or just divide:

$$\frac{45}{250} = 0.18 = \frac{18}{100}$$

$\boxed{18\%}$

7. 6 cracked tiles is 3% of the shipment — how many tiles?

Small percent, small part, missing whole:

$$\frac{6}{n} = \frac{3}{100} \quad \Rightarrow \quad 3n = 600 \quad \Rightarrow \quad n = 200$$

Sanity check on size: 3% is a tiny slice, so the whole must be far bigger than 6 — and 200 is. Check: $0.03 \times 200 = 6$ ✓. $\boxed{n = 200 \text{ tiles}}$

8. Sales rose from 250 loaves to 300 — what percent increase?

For a percent *change*, the part is the amount of change and the whole is always the **original** amount. Change = 300 − 250 = 50; original = 250.

$$\frac{50}{250} = \frac{20}{100}$$

$\boxed{20\% \text{ increase}}$

Common wrong move: using the new amount, 300, as the whole. That gives about 16.7%, which answers a different question.

9. 5.40 dollars of tax on an 80-dollar purchase — what rate?

The tax is the part and the purchase is the whole:

$$\frac{5.40}{80} = 0.0675 = \frac{6.75}{100}$$

A tax rate below 10% should come out as a small percent; 6.75 is reasonable. $\boxed{6.75\%}$

10. 155 yes votes is 62% of the voters — how many voted?

Part = 155, percent = 62, whole missing:

$$\frac{155}{n} = \frac{62}{100} \quad \Rightarrow \quad 62n = 15500 \quad \Rightarrow \quad n = 250$$

Check: $0.62 \times 250 = 155$ ✓. Note the whole is larger than the part, as it must be whenever the percent is under 100. $\boxed{n = 250 \text{ students}}$

11. Price fell from 96 dollars to 72 — what percent decrease?

Percent change again, so the whole is the original price, 96. Change $= 96 - 72 = 24$.

$$\frac{24}{96} = \frac{1}{4} = \frac{25}{100}$$

$\boxed{25\% \text{ decrease}}$

Note: raising 72 back to 96 would be a $24/72 = 33.3\%$ increase, not 25% — the two percents differ because they are measured against different wholes.

12. After a 20% increase the gym has 168 members — how many before?

This is the synthesis problem: 168 is *not* the part that was added. Last year's membership is 100% of itself, and this year is that plus 20% more, so 168 corresponds to 120%.

$$\frac{168}{n} = \frac{120}{100} \quad \Rightarrow \quad 120n = 16800 \quad \Rightarrow \quad n = 140$$

Check: 20% of 140 is 28, and $140 + 28 = 168$ ✓. $\boxed{n = 140 \text{ members}}$

Common wrong move: taking 20% of 168 and subtracting, which gives 134.4 — that removes 20% of the *new* total instead of undoing 20% of the old one.